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PROJECT AND TEAM INFORMATION

Project Title

SkillArc: A Gamified Learning Platform for Data Structures and Object-Oriented Programming

Team Name: *Team #: To be assigned by mentor*	*Apex*
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PHASE-III PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

SkillArc is gamified learning platform designed tohelp computer science student study DataStructures and Algorithms(DSA) in C and ObjectOrientedProgramming(OOP) in C++ The main idea is to bring learning material,topic roadmaps,quizzes,coding practice and progress related features together instead of making student use separate resources for every activity Students can move through topics in planned order,read learning material,practise questions and check their progress. InPhaseIII,we focused on improving both the learning content and the working features of the platform. We added and tested datastructure based features such as Stack navigation,Queuebased study tasks, a Linked List activity feed,Trie auto complete and a MinHeap leaderboard.A graph is used to show the prerequisites between different topicswith BFS and DFS helping students explore their learning paths. We also developed OOP-based user models, added password protectionand expanded the quiz question bank.overallthese features are intended to make selflearning more organised,interactiveand enjoyable for students.

Updated Project Approach and Architecture (2 pts)

The program is create using Python and Streamlit Streamlit helps in creating the userinterface whereas the functionality of the app is done using Python programing lang.The application records are stored using JSONfiles.The entire platform is based on learning modules for DSA in C and OOPinC++.This way students have access to different topics within one app. Each data structure has a unique purpose. The MinHeap is useful for creating a leaderboard based on ranking,which helps to arrange students based on their score The Trie helps in completin the name of a topic through auto complete. Thisfeature allows users to type the beginning of a topic name and complete it Each data structure has a specific role. The MinHeap supports leaderboard ranking,where student score are arranged so that the highestranked students can be displayed. The Trie supports prefix-basedtopic auto complete helping users find a topic by entering only the beginning of its name.A Graph represents the prerequisite relationship between topics.BFS is used to find a shortest route through the topic graph while DFS is used to explore the graph and produce at opological learning sequence. A Stacks to res navigation hi story and follows the Lastin Firstout(LIFO)rule when returning to a previous page.A Queue manages study tasks in Firstin Firstout(FIFO)order.The Linked List stores activity feed entries with new entries inserted at the head. The OOP part uses userStudent User and AdminUser classes to represent commonand role specificuser information. Inheritance and polymorphism support thedifferent user types. Passwords are stored as PBKDF2-HMAC-SHA256 hashes with individual salts and the migration process updates older records The quiz bank stores questions alongwith explanations.



Tasks Completed (7 pts)

Task Completed	Team Member
Implemented and tested multiple DSA features and connected them with platform activities. This included Stack-based previous-page navigation, Queue-based study task handling, Linked List activity feed, Graph-based topic prerequisites, BFS and DFS operations, Trie-based topic autocomplete and Min-Heap leaderboard ranking. The modules were also checked through the Phase III verification script.	Samiksha Gurung (Team Lead and DSA Specialist), with team support.
Worked on the OOP learning part and user-model structure. This includes the User, StudentUser and AdminUser model classes and the use of inheritance and polymorphism to represent different kinds of users.	Tanisha Rai
Contributed to the application interface and integration of learning modules and project features. This helped bring the separate parts of the platform together in a form that students can use.	Vaibhavi Thapliyal

Challenges/Roadblocks (7 pts)

One challenge was connecting different data structures with actual platform features. It was important that each structure should have a clear purpose instead of being included only for theory. We handled this by implementing the features separately and checking their expected behaviour. For example, we checked the Stack for previous-page navigation, the Queue for study-task order and the Linked List for activity feed insertion and traversal.

Another challenge was representing topic prerequisites correctly. A topic graph needs valid connections so that BFS and DFS can explore it in a meaningful order. We tested the sample learning path and the generated topic sequence to check the graph operations.

Managing student records was another area that needed care. Since records are stored in JSON, the application must read and update the information in a consistent way. We added a migration process for older password records and checked that six records were migrated successfully.

Password safety was also important. Instead of storing plain-text passwords, the project uses PBKDF2-HMAC-SHA256 hashing with an individual salt for each record. Verification is performed against the stored hash.

Finally, testing every feature manually would take time as the project grows. We used the automated script ‘verify$_phase3.py'totestthemainmodulestogetherandcheckforanyerrorsHowever, thefinalreviewoftheoveralluse$

Project Outcome/Deliverables (2 pts)

The key achievement in Phase III is an advanced SkillArc platform with the integration of DSA training using C and OOP training using C++ This new version of the platform comprises number of features that illustrate how data structures areapplied in practical applications. for example minHeap facilitates student leaderboard ranking, while Trie autocomplete helps in searching for topics based on a certain prefix. Graph topic illustrates prerequisite requirements and BFS shortest path searching and DFS. Moreover, the following structures are included in the application as well as Stack-based previous page navigation, Queue-based management of study tasks and linked list activity feed. The OOP user models used are User StudentUser and AdminUser. Passwords are encrypted via salted PBKDF2-HMAC-SHA256 hashing. finally the quiz bank comprises 17 modulesand 57 questions with explanation. Thus all of these achievements have improved the learningand tracking process in SkillArc.

Progress Overview (2 pts)

During Phase III the team focused on adding practical features to the platform and connecting them with the existing learning modules. The work covered the main DSA components, including StackQueue, linked list, Trie, Min Heap and Graph operations. BFS and DFS were checked for topic path operations. The OOP user models and password handling process were also included in the verification The quiz bank was checked for the expected number of modules, questions and explanations. We ran the automated Phase III verification script across the main features. All 10 test groups passed successfully with zero errors.The tests covered password hashingJSON migration, leaderboard ranking, autocomplete, graph traversal, navigation, study-task handling, activity feed operations, OOP user models and the quiz bank. The main development work covered by these tests is complete.The remaining work is a final endto end review of the application, checking that the modules are easy to navigate and that the displayed information is understandable. Any small issues found during this review will be corrected before submission. We have not assigned numerical completion percentage because this final review is still pending.

Codebase Information (2 pts)

GitHub Repository: `https://github.com/samiksha1897-bot/SkillArc-PBL`
Branch: `main`
The project source code and phase-wise updates are maintained in the GitHub repository. The main branch contains the current integrated version of the project. GitHub is used to keep the project files in one shared place, maintain the commit history and allow team members to work on and review updates.
Important Phase III commit: `Implemented Phase 3 modules and features`. This commit records the addition of the Phase III modules and features. The repository link can be used to view the source files and the available commit history.

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
Password hashing	PASS	PBKDF2-HMAC-SHA256 and verification passed.
JSON record migration	PASS	6 records migrated with individual salts.
Min-Heap leaderboard	PASS	9 students ranked correctly.
Trie autocomplete	PASS	Prefix searches for link, inord and virt passed.
Graph BFS DFS	PASS	verified that all prerequisites andtopics are arranged in the correct order
Navigation Stack	PASS	previous-page navigation feature worked as expected
Study task Queue	PASS	ffirst task executed first
Linked List activity feed	PASS	insertion at the beginning of the linkedlist and its traversal were tested successfuly
OOP user models	PASS	User, StudentUser and AdminUser verified.
Quiz question bank	PASS	17 modules 57 questions verify

Deliverables Progress (2 pts)

Completed: Phase III includes DSA and OOP learning materials and new features such as stack-based navigationn queue-based study task management and a linkedlist activity feed.it also covers graph representation of topic prerequisites, BFS and DFS traversal, Trie-basedautocomplete and a Min-Heap leaderboard.the user management system includes User, StudentUser and AdminUser modelsFor better password security PBKDF2-HMAC-SHA256 hashing with individual salts has been implemented, along with JSON password-record migration. The quiz bank has also been expanded to include 17 modules 57 questions. The automated verification script also completed successfully with all 10 test groups passing.
In progress / final check: A complete end-to-end review of the application, navigation and displayed content.
Pending before submission: Apply any small corrections found during the final review and complete the final documentation check.